Marketing Campaign Customer Segmentation

Technical Portfolio Project & Repository Documentation

📌 Project Overview

This project establishes an end-to-end data engineering and unsupervised machine learning pipeline designed to execute robust, algorithmic customer segmentation. Operating on an enterprise dataset encompassing over 22,000 distinct customer interactions, the pipeline automates data cleansing, engineers behavior-driven demographic indicators, and builds iterative clustering models. The objective is to translate static, multi-dimensional consumer interactions into actionable consumer archetypes, enabling automated, hyper-targeted promotional campaigns.

🛠️ Technical Infrastructure & Tech Stack

Pipeline Component	Technologies & Frameworks Utilized
Data Engineering & Processing	Python, NumPy, Pandas
Unsupervised Machine Learning	Scikit-Learn (LabelEncoder, StandardScaler, PCA, KMeans, AgglomerativeClustering)
Cluster Optimization & Validation	Yellowbrick Metrics (KElbowVisualizer)
Data Visualization Engine	Matplotlib (Pyplot, ListedColormap, Axes3D), Seaborn

🏗️ Pipeline Architecture & Implementation Walkthrough

1. Exploratory Data Analysis (EDA) & Data Cleansing

- **Socioeconomic Missing Value Treatment:** Identified and addressed sparsity within structural fields (such as missing `Income` logs) via deterministic row validation and deletion to prevent calculation bias.
- **Temporal Feature Transformation:** Parsed raw timestamps (`Dt_Customer`) against active dates to synthesize a continuous metric for long-term customer tenure (`Customer_For`).

2. Feature Engineering & Scaling

- **Metric Synthesis:** Aggregated multi-column spending habits across parallel product lines (Wines, Fruits, Meat, Fish, Sweets, Gold Products) to construct a comprehensive fiscal value metric (`Spent`).
- **Demographic Categorization:** Distilled granular, multi-class categorical entries regarding relationship status and academic backgrounds into unified behavioral attributes (`Living_With`, `Family_Size`, and `Is_Parent`).

- **Feature Normalization:** Configured a `StandardScaler` routine to uniformly distribute data variances, guaranteeing distance-based metrics are not dominated by differences in units.

3. Dimensionality Reduction

- **Principal Component Analysis (PCA):** Executed linear dimensionality reduction to transform collinear data matrices into orthogonal principal components. This effectively compressed the feature space, minimized computing overhead, and isolated the major directional vectors driving customer behavior variance.

4. Iterative Clustering Optimization

- **K-Means Iteration:** Deployed centroid-based K-Means processing linked directly to a `KElbowVisualizer` to dynamically parse distortion scores and locate the mathematically optimized cluster quantity (K).
- **Hierarchical Cross-Validation:** Validated the K-Means cluster topologies using bottom-up `AgglomerativeHierarchicalClustering`, verifying the boundaries, density, and stability of the derived segments.

📊 Repository Architecture & Directory Structure

```
├── data/
│   └── marketing_campaign.csv      # Cleaned and processed raw dataset
├── notebooks/
│   └── customer_segmentation.ipynb # Full pipeline walkthrough and exploratory
analysis
├── src/
│   ├── data_preprocessing.py       # Modular source code for cleansing and features
│   ├── clustering.py               # PCA algorithms and clustering models
│   └── visualization.py            # Scripts executing 3D vector representations
├── requirements.txt                # System and package dependencies
└── README.md                       # Repository profile and deployment blueprint
```

📈 Enterprise Impact & Actionable Insights

> **Strategic Business Outcomes:**
> • **Ad-Spend Optimization:** Transformed broad legacy ad strategies into precise, segment-driven initiatives, lowering budget waste.
> • **Demographic & Household Mapping:** Isolated high-yield non-parent vs. parent behavior archetypes, allowing automated marketing engines to tailor campaign copy based on household size and purchasing bandwidth.
> • **Retention Management:** Leveraged `Customer_For` tenure data to map client lifecycles, enabling early identification of decaying recency metrics to trigger automated win-back workflows.